

Portfolio Summary

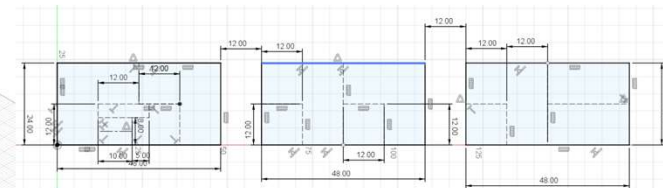
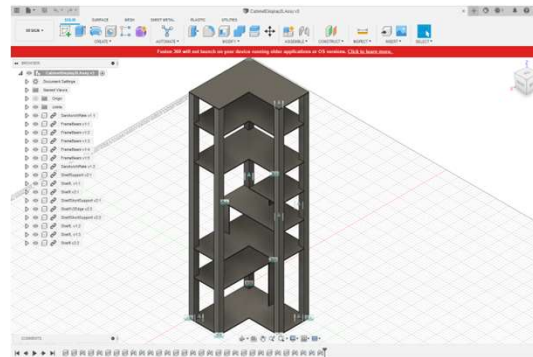
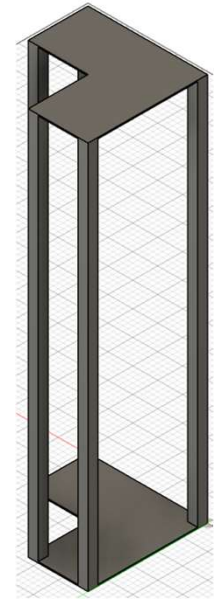
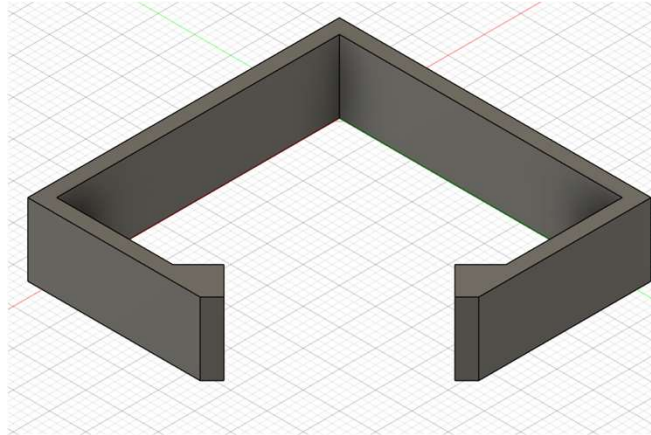
Henry Luo

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Pine Display Shelf:

- Objective:
 - Create an L-shaped wooden display shelf to fit in a uniquely-shaped room
- Criteria & Challenges
 - Only have access to a jigsaw and a drill (DFM/DFA)
 - Must spend less than \$100
 - Should hold reasonably heavy items sturdily
 - Aesthetically pleasing to client



Pine Display Shelf Construction



Day 0: Raw Materials



Day 1: Plywood Cut



Day 2: Frame Completion



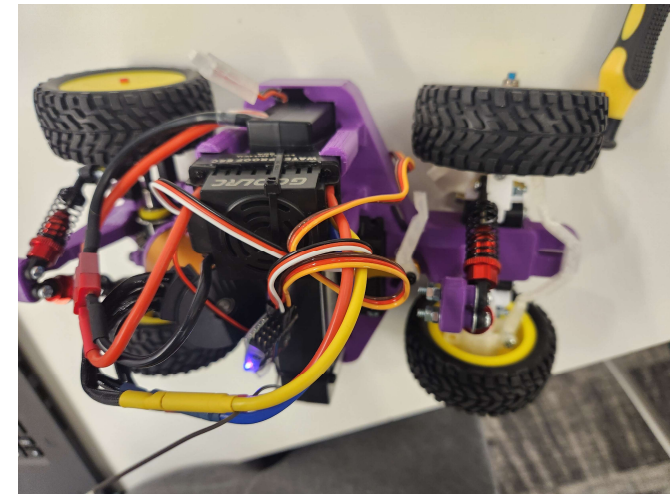
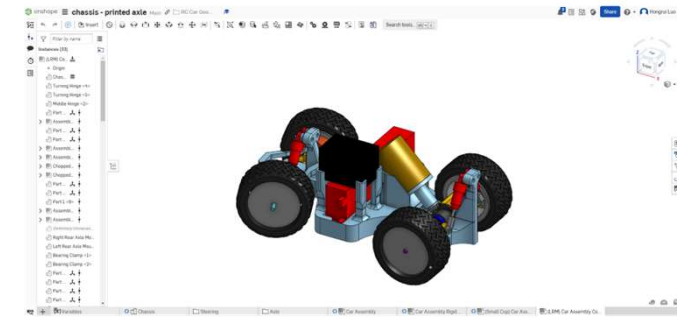
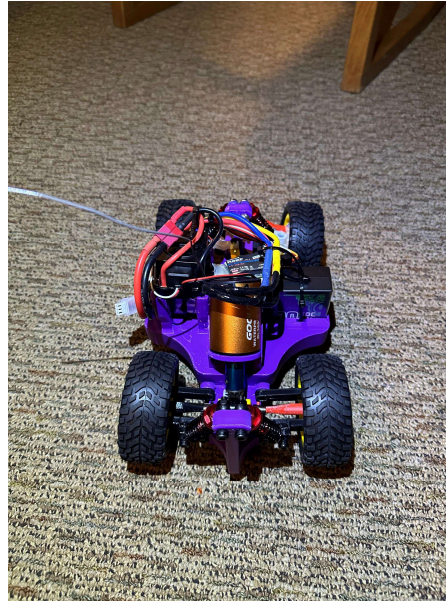
Day 2 End: Shelf Completion



Pine Display Shelf

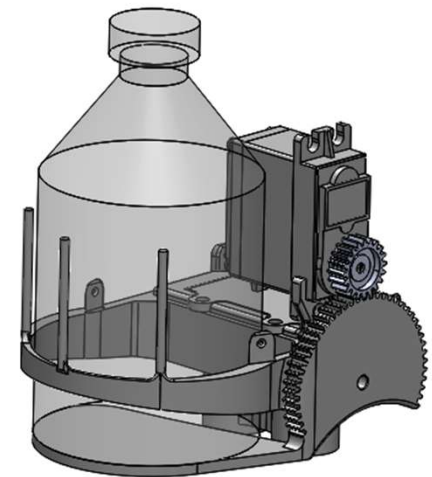
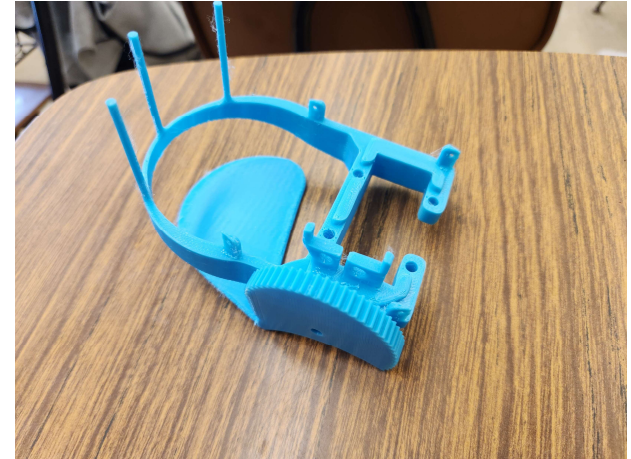
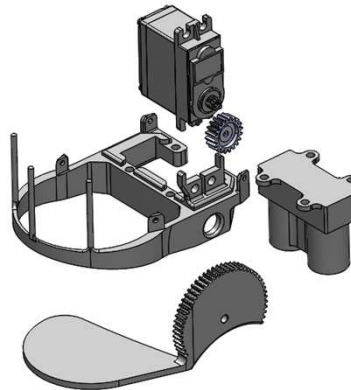
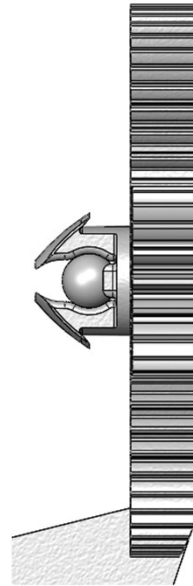
ASME Mini RC Car:

- Objective:
 - Create a mini RC car in 12 weeks to win a race with jumps, bumps, turns, and straights
- Criteria & Challenges
 - Must fit in a 6"x10" rectangle
 - Must spend less than \$125
 - Randomly-assigned team



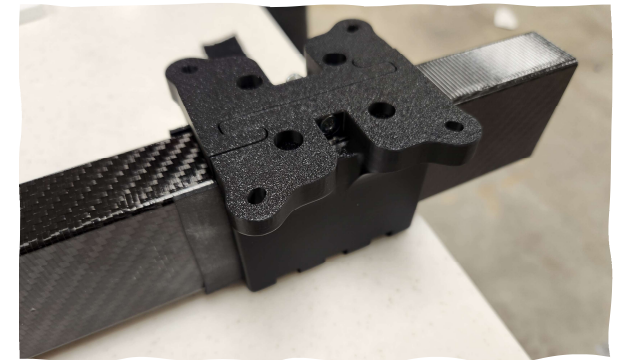
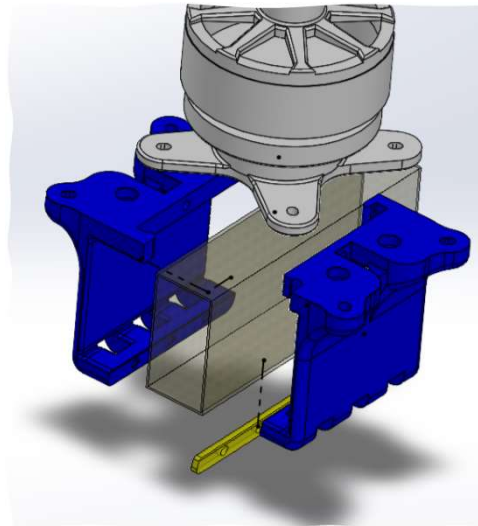
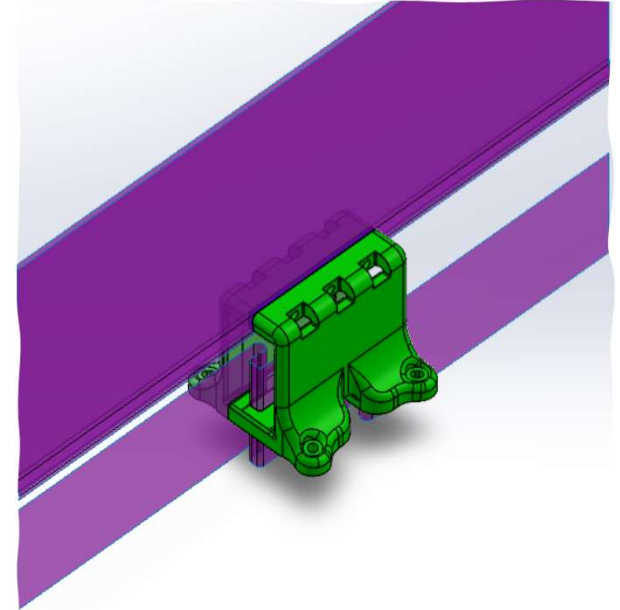
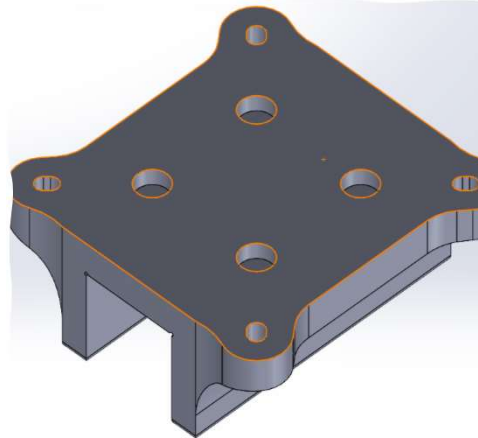
PART Bachelor Payload Capsule:

- Objective:
 - Create a payload capsule to release a water bottle payload when close to the ground
- Criteria & Challenges
 - Must be 3D printed
 - Must share electronic components with the sister "trinity" design
 - Should be as light as possible



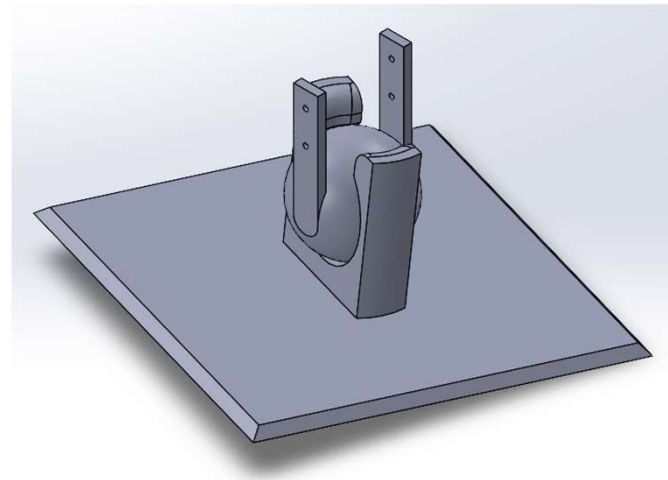
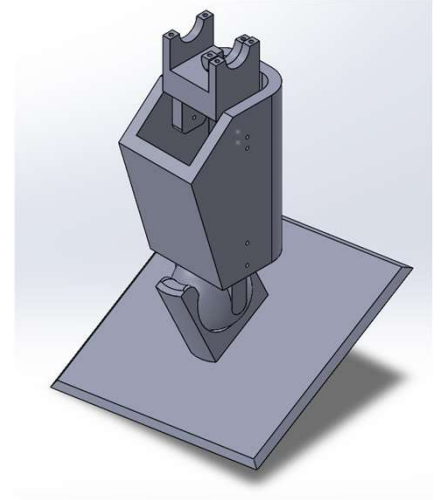
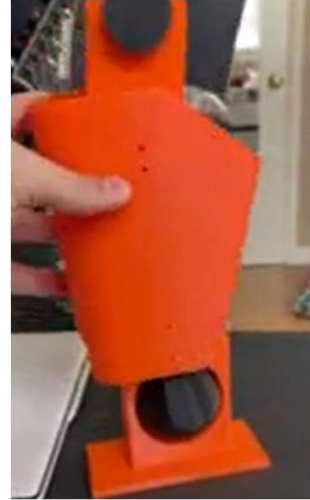
PART Rear Motor Mount:

- Objective:
 - Modify an existing rear motor mount design to eliminate composite tail boom delamination concerns
- Criteria & Challenges
 - Must be secure
 - Must not change the tail boom structure
 - Must secure the rear motor



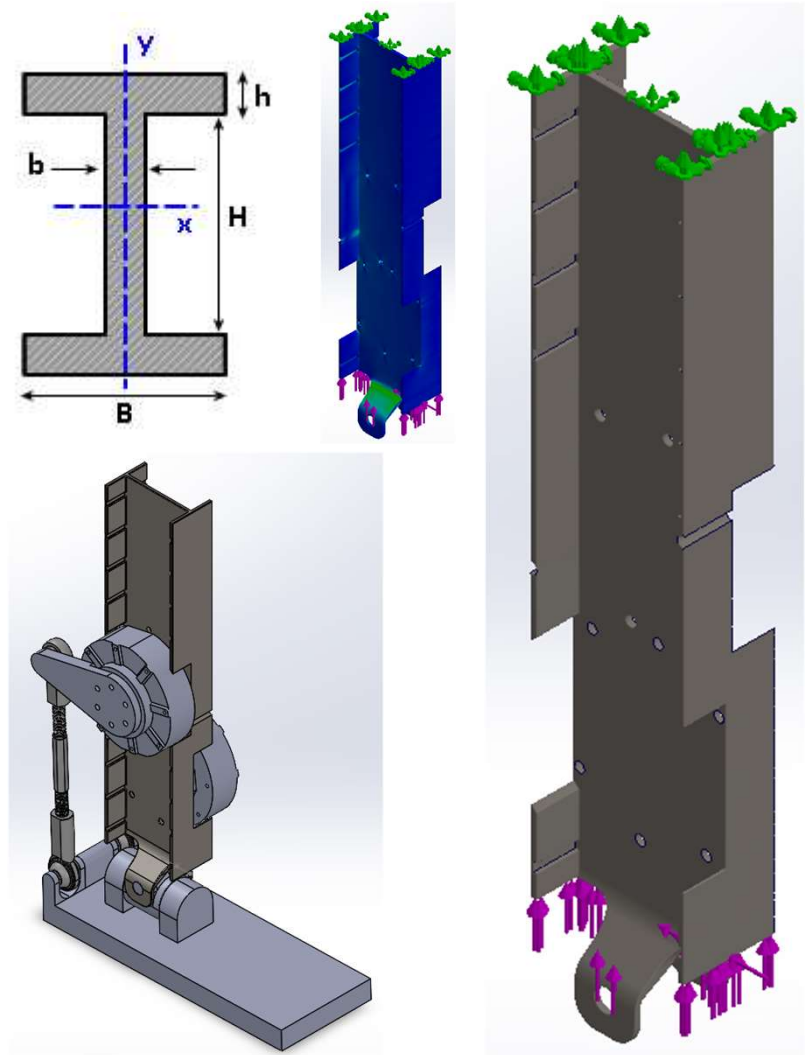
HRC Balljoint:

- Objective:
 - Create a jig to simulate ankle movements for the development of ankle controls
- Criteria & Challenges
 - Must be easy to assemble
 - Must be easy to use (stays in socket)
 - Must accurately represent ankle rolling movement



HRC Shin/Calf:

- Objective:
 - Create a full-scale structural shin
- Criteria & Challenges
 - Must absorb axial compressive loads
 - Must transfer shear (forward/backward) loads without bending
 - Must minimize stress concentrations using fillets where possible





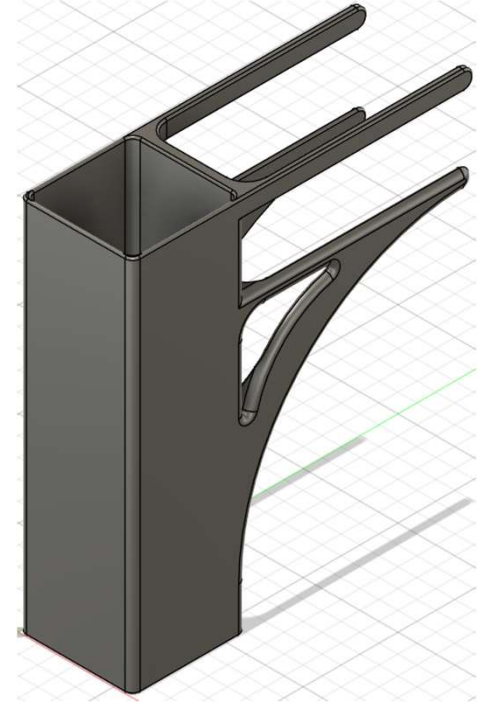
Clip-On Sunglasses Holder



Display Hooks



Clothes Hanger



Desk Pencil Holder

Miscellaneous 3D Printed Projects:

Developed Skills:

- Hard Skills:

- CAD
- FEA
- Woodworking
- 3D Printing

- Soft Skills:

- Product Development
- DFM/DFA
- FTR vs Iterative Improvement
- Leadership/Mentorship

Thank You!

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